Aerostat GCS Dashboard
End-to-End Data Process

Document for Management Review

Purpose

This document describes the complete end-to-end **data process** of the Aerostat Ground Control Station (GCS) software — from device connection to live monitoring, logging, and report export.

End-to-End Process Overview

Step	Stage	Input	Output
1	Connect device	IP/Port or COM settings	Live data link
2	Parse & map data	Raw CSV from device	Named telemetry
3	Display live	Mapped parameters	Real-time dashboard
4	Monitor alerts	Threshold limits	Warnings if out of range
5	Log automatically	All updates	data.json audit file
6	Export reports	Date range + format	CSV / JSON / TXT file
7	Deploy / reuse	Windows executable	Field-ready GCS tool

Step 1 — Connect to the Device

Action	Detail
Launch software	Run dashboard on GCS laptop (AerostateDashboard.exe)
Choose connection	Ethernet/TCP (IP + Port) or COM/Serial (Port + Baud rate)
Connect	Admin → Device → Connect; status shows Connected, lines received
Verify	Dashboard parameters begin updating within seconds

Step 2 — Receive & Process Device Data

The device sends CSV sensor data over Ethernet or COM. The dashboard automatically parses, maps, and stores each update for display and logging.

Category	Parameters received from device
Environmental	Ambient temperature, ambient pressure, humidity
Pressures	Helium pressure, pressure difference (ΔP)

Category	Parameters received from device
Temperatures	Helium temperature, ground (DHT) temperature
Position	Altitude (AMSL & AGL), latitude, longitude, pitch, roll, heading, compass
Wind & Tether	Wind speed, confluence point tension (C.P. Tension)
System	Ping / response time

Step 3 — Live Monitoring on Dashboard

Capability	Description
Live values	All connected parameters update automatically on screen
Organized sections	Atmospheric, Pressures, Temperatures, Position, Wind, Tether
Charts	Trend graphs for key parameters over time
Timestamp	Last update time visible for operational awareness
Admin control	Admin can choose which parameters operators see

Step 4 — Alerts & Threshold Monitoring

Action	Detail
Set limits	Admin defines min/max thresholds per parameter
Monitor	System continuously compares live values to limits
Alert	Notification shown when a parameter goes out of range
Smart alerts	Only visible parameters receiving live device data trigger alerts

Step 5 — Automatic Data Logging

Every sensor update is saved to **data.json** with date, time, source (ethernet or COM), all parameter values, and change details. Logging runs continuously while the device is connected.

Step 6 — Export & Reporting

Action	Detail
Open Logs	Use Logs button on dashboard; select date range
Choose format	CSV, JSON, or TXT
Download	File ready for Excel, analysis, or archival
Source tagging	Each record tagged as ethernet or com:COMx

What Management Gets

Benefit	Description
Single platform	One software for connect → monitor → log → export
No manual scripts	Operators use dashboard UI only
Traceability	Full history with timestamps and data source
Field deployment	Windows .exe — deploy without developer support
Reporting ready	Export data for reviews and records

Summary: We have developed an end-to-end GCS data system. The operator connects the aerostat device over Ethernet or COM, the dashboard displays live telemetry with alerts, all data is logged automatically, and reports can be exported for management and analysis — delivered as deployable Windows software.

Scope: Data acquisition, live display, alerting, logging, and export.